

# Secure Network Design

A "Secure Network Design" refers to the process of planning and building a computer network with a primary focus on implementing robust security measures, aiming to protect data and systems from unauthorized access, malicious attacks, and data breaches by incorporating various security technologies, access controls, and best practices throughout the network architecture, essentially designing a network with security at its core.

## Key aspects of Secure Network Design:

### Risk Assessment:

Identifying potential threats and vulnerabilities within the network to prioritize security measures based on the level of risk associated with sensitive data. 

### Defense in Depth:

Implementing multiple layers of security controls to prevent attackers from easily penetrating the network, including firewalls, intrusion detection systems, and strong authentication methods. 

### Network Segmentation:

Dividing the network into smaller, isolated segments to limit the impact of a breach and prevent lateral movement of attackers. 

### Least Privilege Principle:

Granting users only the minimum level of access needed to perform their job functions, reducing potential damage from compromised accounts. 

### Strong Authentication:

Utilizing multi-factor authentication, strong passwords, and secure protocols to verify user identity and prevent unauthorized access. 

### Data Encryption:

Encrypting sensitive data both at rest and in transit to protect it from unauthorized access even if intercepted. 

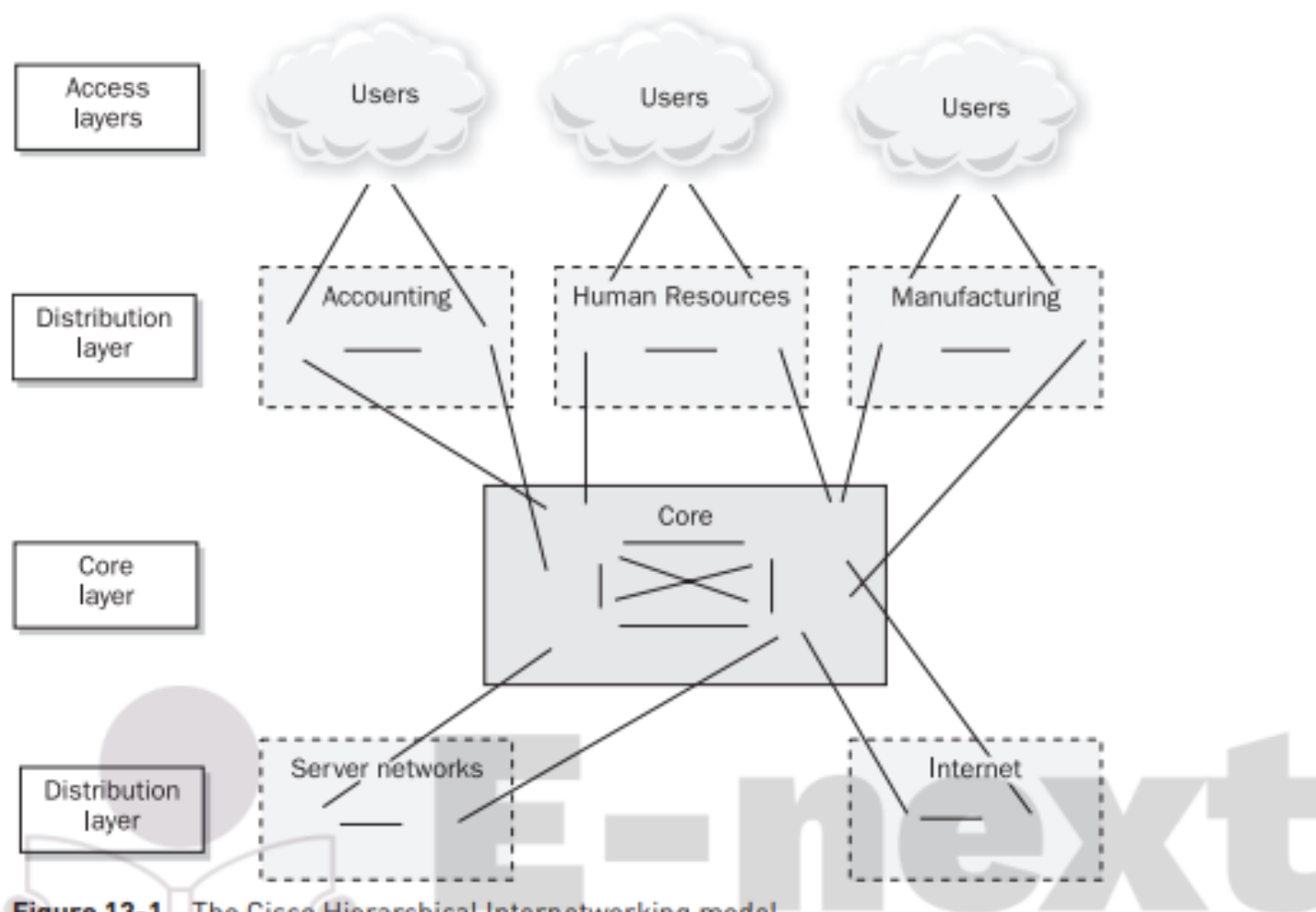
### Vulnerability Management:

Regularly scanning for and patching known vulnerabilities in network devices and software to mitigate potential threats. 

# Performance

The network will play a huge role in meeting the performance requirements of an organization. Networks are getting faster and faster, evolving from 10 megabit to 100 megabit to gigabit speeds, with 10GE commonly deployed and 40GE, 100GE, and InfiniBand technologies available today. When determining the appropriate network technology, be sure that it can meet the bandwidth requirements projected for three to five years in the future. Otherwise, expensive replacements or upgrades may be required.

Applications and networks that have low tolerance for latency, such as those supporting video and voice streaming, will obviously require higher performance network connections and hardware. What about applications that move data in large chunks (for example, storage snapshots or disk-to-disk offsite replication)? In lieu of an expensive, dedicated, high-bandwidth connection, it may be more economical to implement links that are *burstable*, meaning that the provider will allow short bursts of traffic above the normal subscribed rate. If applications will share common network infrastructure components, the design team may also consider implementing Quality of Service (QoS) technologies to prevent one application from consuming too much bandwidth, or to ensure that higher priority applications always have sufficient bandwidth available.



**Figure 13-1** The Cisco Hierarchical Internetworking model

- **Distribution layer** Sits between the core and the access layer. This layer is used to aggregate access-layer traffic for transmission into and out of the core.
- **Access layer** Composed of the user networking connections.

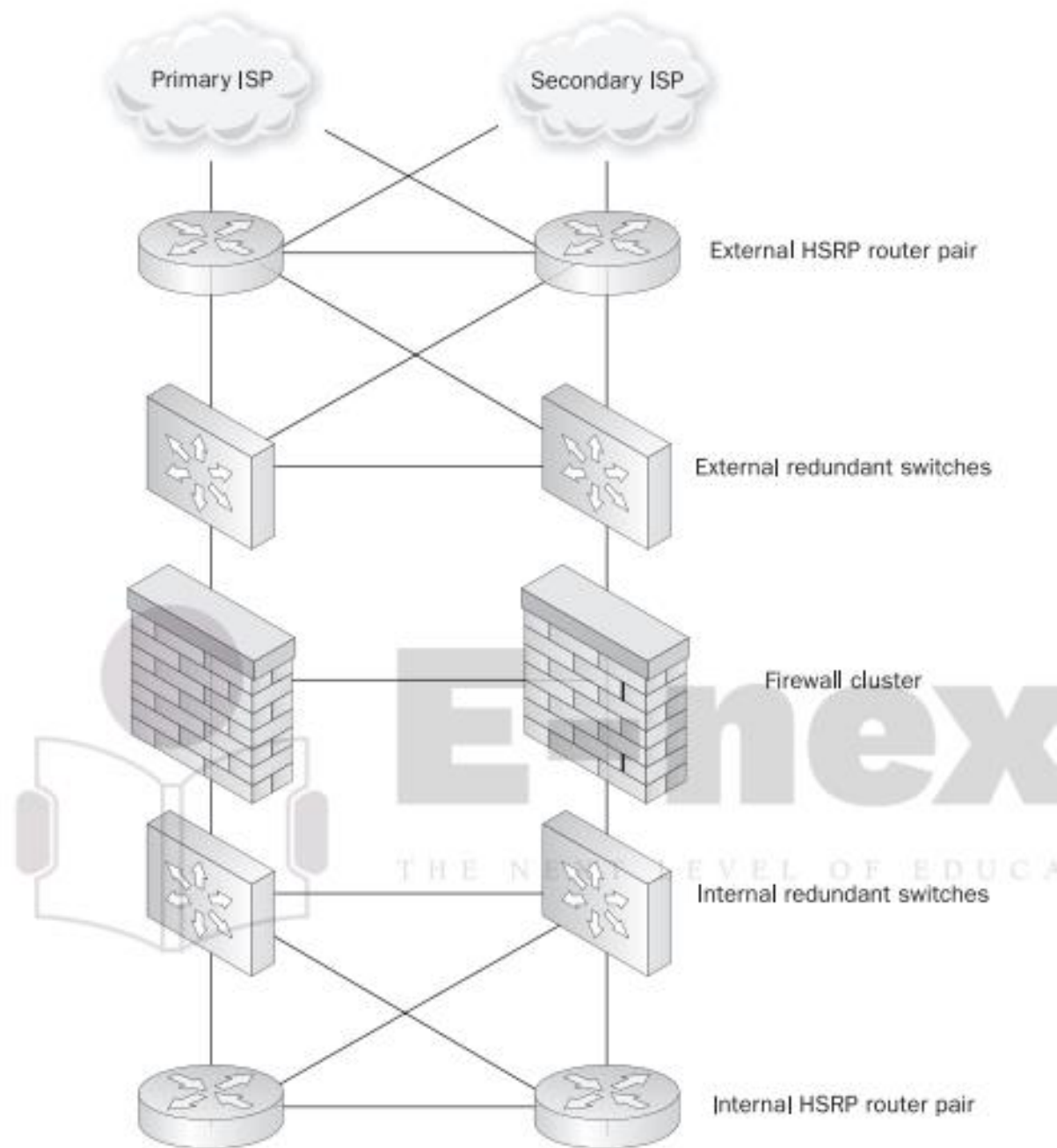
# Availability

Network availability requires that systems are appropriately resilient and available to users on a timely basis (meaning, when users require them). The opposite of availability is denial of service, which is when users cannot access the resources they need on a timely basis. Denial of service can be intentional (for example, the act of malicious individuals) or accidental (such as when hardware or software fails). Unavailable systems cost corporations real dollars in lost revenue and employee productivity, and they can hurt organizations in intangible ways through lost consumer confidence and negative publicity. Business availability needs have driven some organizations to construct duplicate data centers that perform real-time mirroring of systems and data to provide failover and reduce the risk of a natural disaster or terrorist attack destroying their only data center.

Depending on the specific business and risk factors, redundancy often increases both cost and complexity. Determining the right level of availability and redundancy is an important design element, which is best influenced by a balance between business requirements and resource availability.

The best practice for ensuring availability is to avoid single points of failure within the architecture. This can require redundant and/or failover capabilities at the hardware, network, and application functions. A fully redundant solution can be extremely expensive to deploy and maintain, because as the number of failover mechanisms increases, system complexity increases, which alone can raise support costs and complicate troubleshooting.

As mentioned previously, the application's availability requirements should be assessed to determine the financial and business impacts of systems being unavailable. Performing this assessment will help management arrive at the optimal balance between failover mechanisms, cost, and complexity for the particular network or application. Numerous security appliance vendors have failover mechanisms that enable a secondary firewall to assume processing responsibilities in the event that the primary firewall fails. Beyond firewalls, routers can also be deployed in a high-availability configuration.



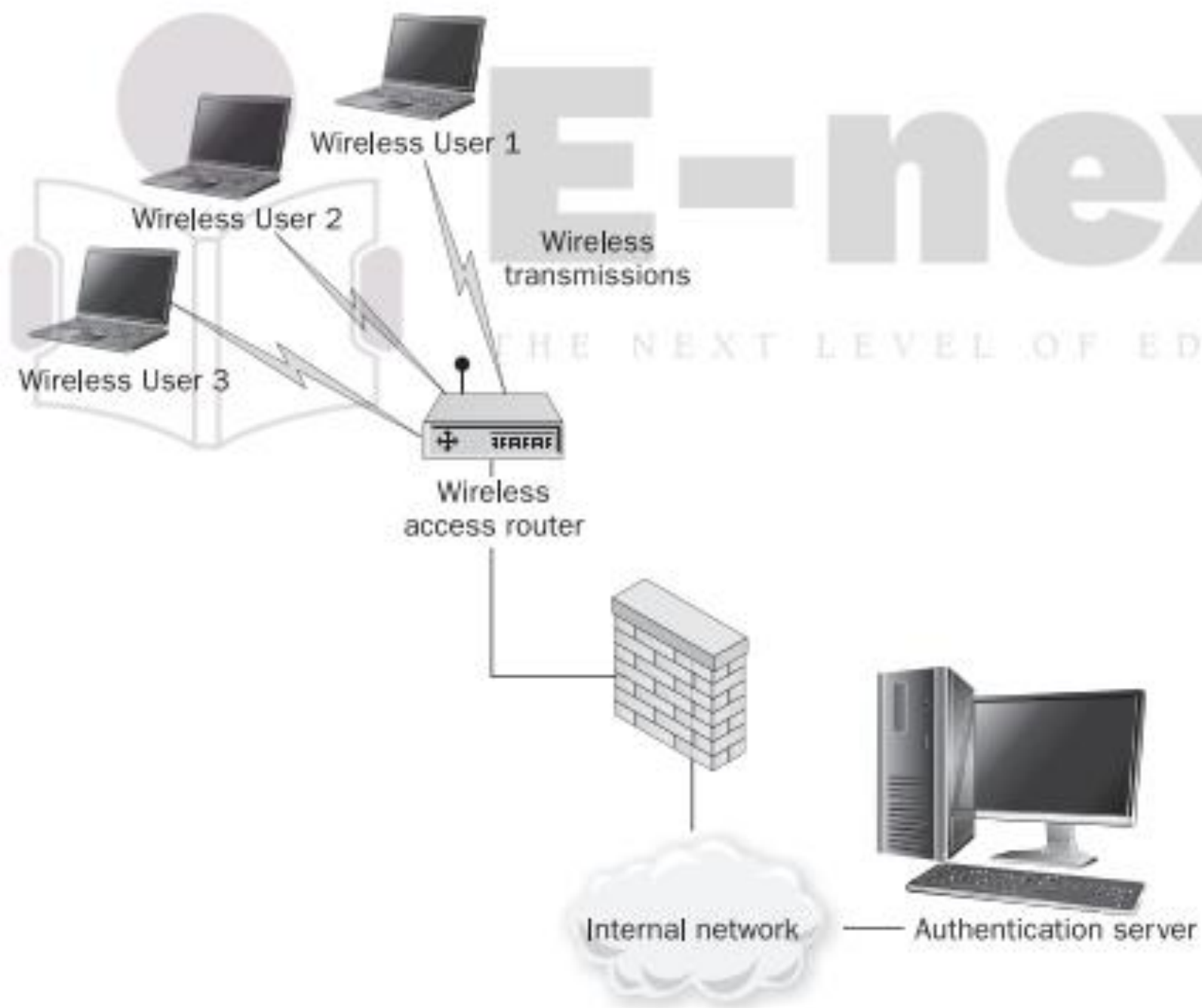
# Security

Each element on a network performs different functions and contains data of differing security requirements. Some devices contain highly sensitive information that could damage an organization if disseminated to unauthorized individuals, such as payroll records, internal memorandums, customer lists, and even internal job-costing documents. Other devices have more exposure due to their location on the network. For example, internal file servers will be protected differently than publicly available web servers.

When designing and implementing security in network and system architectures, it is helpful to identify critical security controls and understand the consequences of a failure in those controls. For example, firewalls protect hosts by limiting what services users can connect to on a given system. Firewalls can allow different sets of users selective access to different services, such as allowing system administrators to access administrative services while preventing non-administrative users from accessing those same services. This provides an additional level of control over that provided by the administrative mechanisms themselves. By denying a non-administrative user the ability to connect to the administrative service, that user is prevented from mounting an attack directly on that service without first circumventing the firewall.

However, simply restricting users to specific services may be insufficient to achieve the desired level of security. For example, it is necessary to allow traffic through the firewall to connect to various authorized services. In order for an organization to send and receive e-mail, firewalls must be configured to permit e-mail traffic. As Chapter 15 will discuss, firewalls have limited capability in preventing attacks directed at authorized applications, so overall network security is dependent on the proper and secure operation of those applications.

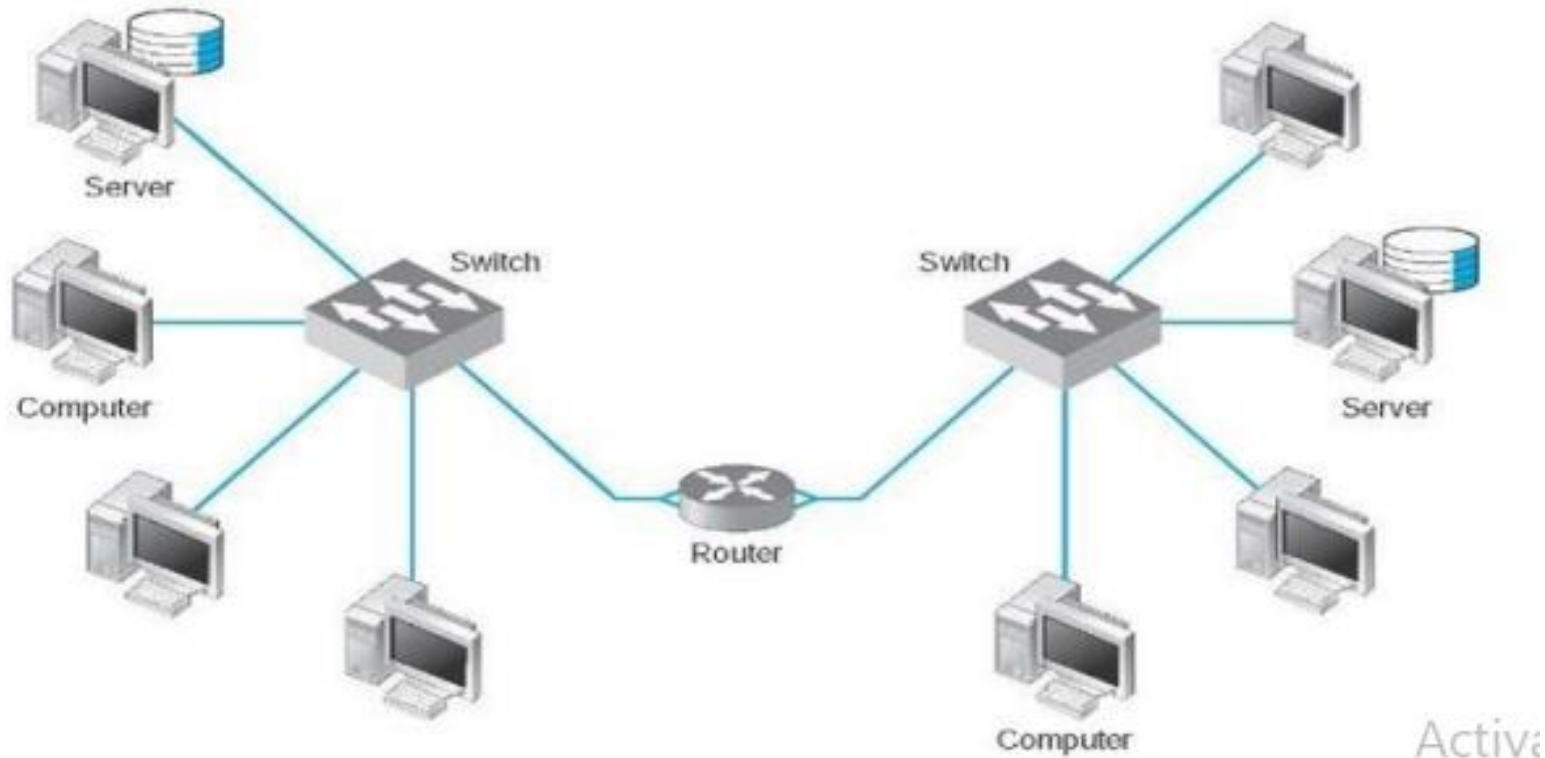




# Network Device Security

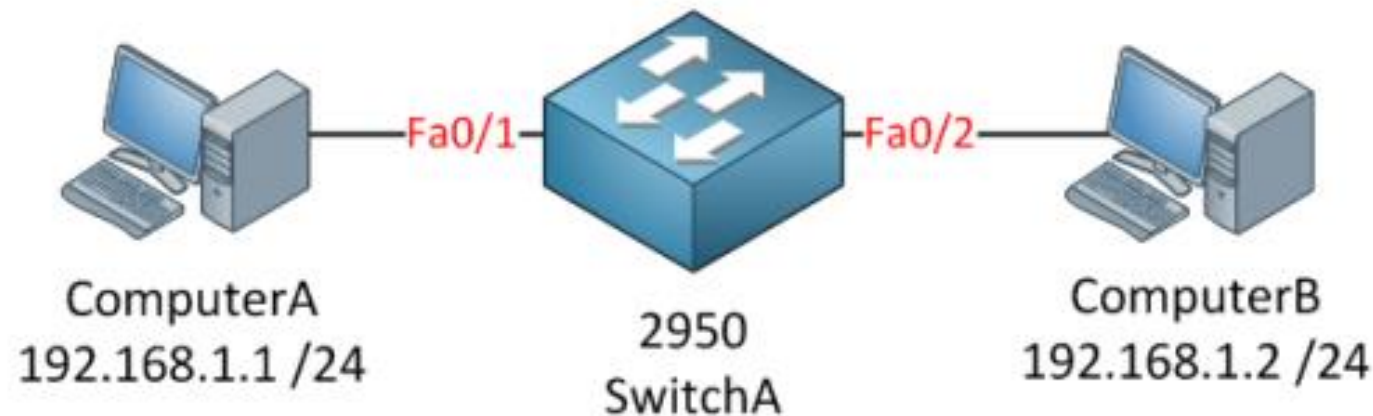
# Router

- **Receive, analyze and move** incoming packets to another networks
- Connecting **different networks**
- Creates and maintains a **routing table of best routes**

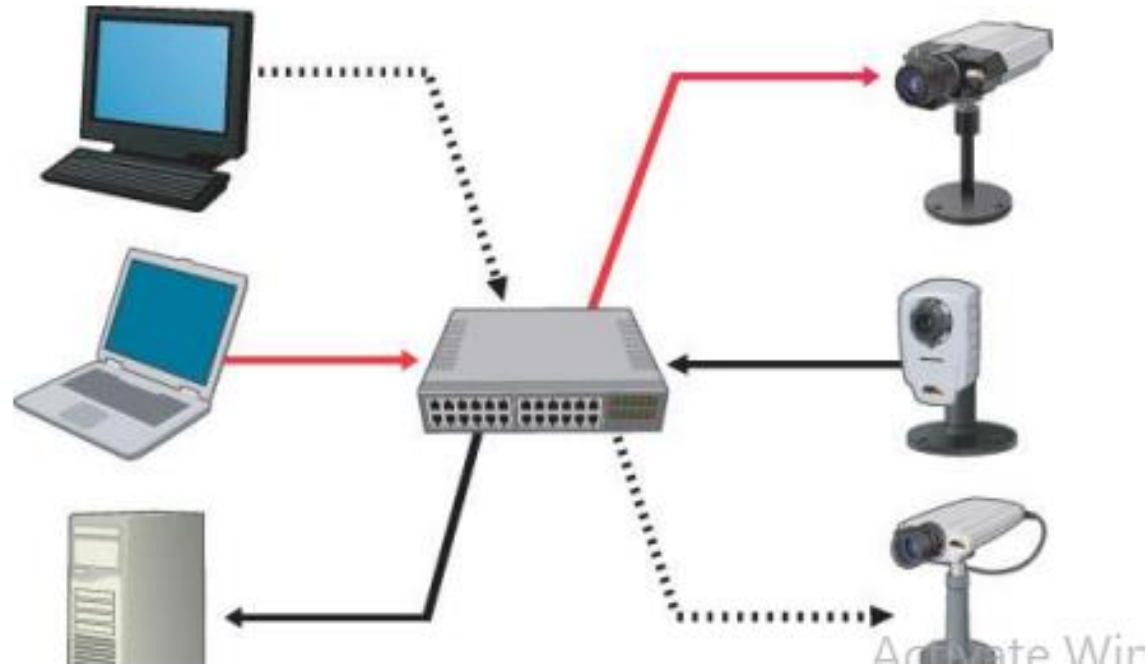


- Used to connect **same networks**
- Having **higher-port density** and performs **forwarding decisions** at wire speed
- Manufactured by using **hardware circuits called ASIC**(Application-Specific Integrated Circuits)
- Builds **MAC-address tables** through a **dynamic learning process**

## Switch Operation

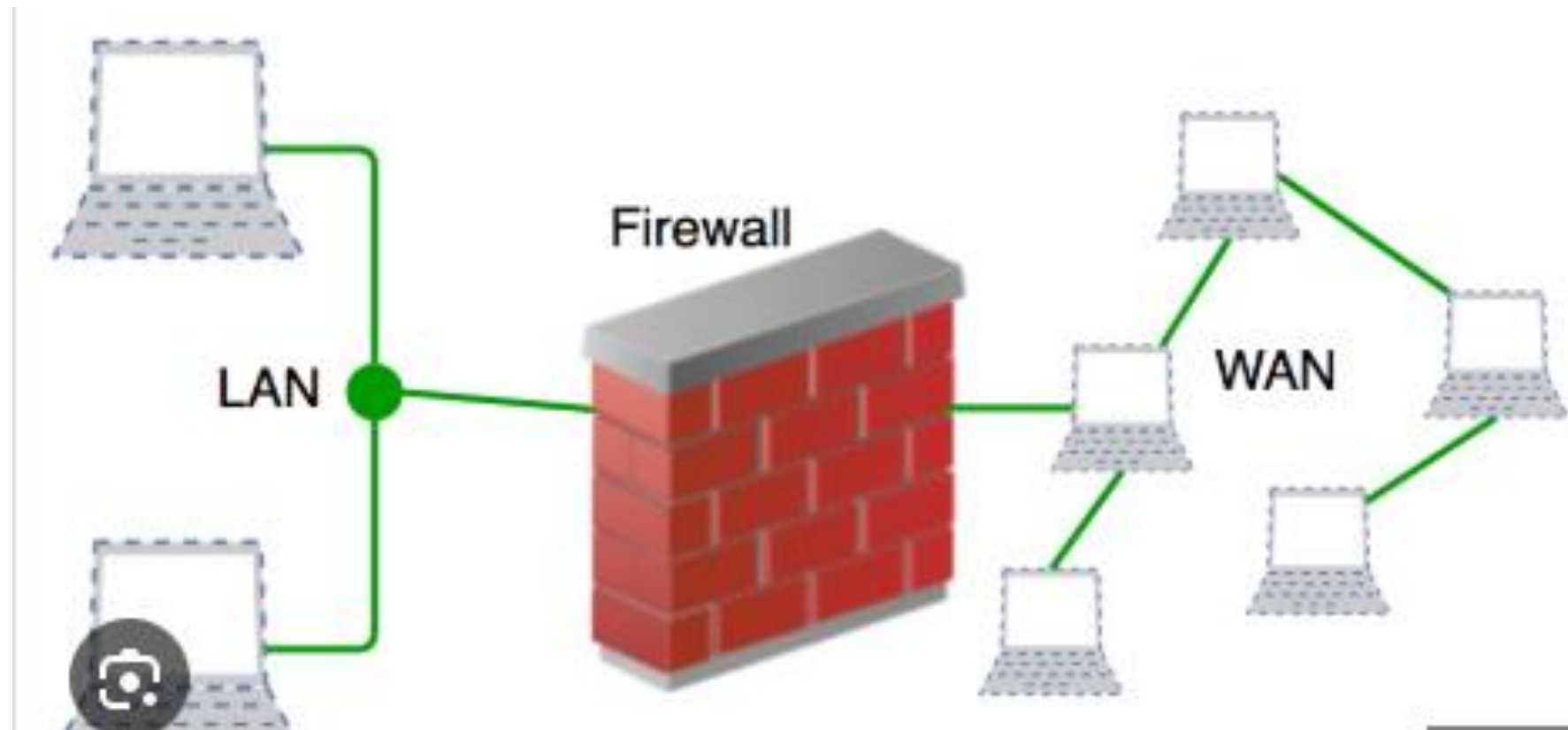


- Switches operate in **full-duplex**
- Switches perform
  - **Hardware address learning**
  - **Intelligent forwarding of frames**
  - **Loop Avoidance**



# Firewall

- A firewall is a network security device that controls incoming and outgoing network traffic. It protects networks from unauthorized access, harmful activities, and potential threats.



## How firewalls work

- Firewalls inspect network packets and decide whether to allow or block them based on pre-set rules. 
- Firewalls can be hardware, software, software-as-a-service (SaaS), or public or private (virtual) cloud. 
- Firewalls can filter traffic based on IP address, ports, protocols, and other factors. 

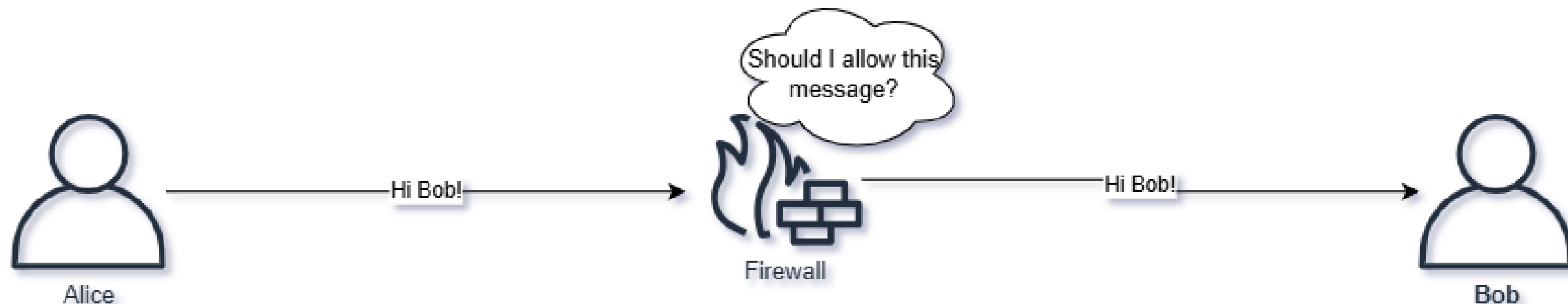


## Types of firewalls

- **Network-based firewall:** Routes traffic between networks 
- **Application firewall:** Filters and intercepts all traffic packets to/from apps 
- **Proxy firewall:** Acts as an intermediary between internal and external networks 
- **Stateful inspection firewall:** Inspects every data packet and compares it against a threat database 
- **Next-generation firewall (NGFW):** Combines traditional firewall technology with other network device filtering functions 
- **Virtual firewall:** Runs entirely within a virtualized environment 

# Segmentation

Firewalls can segment traffic between hosts and systems into segments, sometimes called zones. Each segment holds services which are allowed to communicate between one another.



Any connection to or from the segment should be carefully controlled by the Firewall, preventing any unauthorized connections to make successful connections. Smaller segments offer more segregation, but requires more management.

Without any segmentation, users and systems can talk directly to each other without the Firewalls enforcement. This is called a flat network.

# Wireless network security

- Wireless network security is a part of cybersecurity that protects wireless networks from unauthorized access and breaches. It involves the use of strategies to protect the confidentiality, integrity, and availability of wireless networks.

## How does wireless network security work?

- **Encryption:** Encodes data transmitted over the network to prevent unauthorized access 
- **Protocols:** Uses protocols like Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), and Wi-Fi Protected Access II (WPA2) to secure wireless networks 
- **Virtual Private Networks (VPNs):** Creates secure tunnels between unprotected Wi-Fi networks and the internet 

## Why is wireless network security important? @

- Prevents threats like interception, data theft, and denial-of-service attacks
- Protects the confidentiality, integrity, and availability of wireless networks and their resources

## Examples of wireless network security

- **Wi-Fi Protected Access (WPA):** Uses Temporal Key Integrity Protocol (TKIP) for data encryption @
- **Wi-Fi Protected Access II (WPA2):** Uses a dynamic key that is generated and shared between users on a network @
- **Virtual Private Networks (VPNs):** Uses encryption and identity authentication to allow users to connect to private networks over public internet connections @

# NETWORK SECURITY



**FIREWALLS**



**ENCRYPTION**



**INTRUSION DETECTION  
SYSTEMS (IDS)**



**DATA LOSS  
PREVENTION**



**ANTIVIRUS  
SOFTWARE**



**SECURITY POLICIES**



**SECURE  
COMMUNICATIONS**

